

23 Communication in Sports Teams

Jonas Akpetou and Philip Furley

Chapter Objectives

- The present chapter both applies mainstream theorizing in the field of communication to the sports domain and uses the sports domain to provide novel insights into the science of communication.
- Review and integrate a highly diverse body of literature from the domain of sports within a communication perspective.
- Derive evidence-based recommendations on verbal and nonverbal communication when working in the context of sports.

Introduction

Communication is a central aspect of human nature shaping social life as we know it. Generally speaking, it can be viewed as the process of conveying and receiving information linking different parts of the living world (Littlejohn et al., 2011) and thereby enabling us to interact with other individuals. This process can occur explicitly with the intent to affect others' cognitions and behavior or implicitly as an unconscious display of social information (Littlejohn et al., 2011). Furthermore, information can be transmitted in different forms (e.g., verbally or nonverbally). Either way, the relevance of communication extends to all contexts in which some form of social interaction occurs.

An area characterized by the constant need for communication is the domain of team sports. Sport teams act in the pursuit of a common goal and team members are placed in highly complex and dynamic environments in which they have to interact with regard to the team's objectives and the opponent team's actions (Reimer et al., 2006). Thus, team sports hold numerous demands for efficient coordination, instruction, feedback, and the regulation of the social climate within the group (Ishak, 2017). Those demands can be met through direct communication and effective speech aiming at exchanging intentions and plans accordingly. Besides the open verbal and non-verbal exchange between teammates or rivals, many sports have elaborated symbolic communications (e.g., signs in beach volleyball, or baseball indicating the next play) that are intended to maintain the privacy of the content (e.g., game plans, strategies).

As team athletes are interdependent and have to coordinate their actions in the best possible way, the ongoing simultaneous exchange of information between multiple team members (i.e., group communication) is crucial (Eccles & Tran, 2012). Moreover, regular interpersonal communication, for example, between the coach and specific players, constitutes another important part of the interaction within a team. In addition, the interpersonal communication with reporters and fans, making players and coaches available for interviews, is also promoted.

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Research points out that effective team performance in dynamic settings is facilitated when members can develop adequate expectations regarding their team mates' roles, actions, and decisions (Blickensderfer et al., 2000). This enhances adaptation to changing demands in the environment. To consolidate the mutual understanding of each athletes' roles in the team extensive communication is necessary.

However, when overt communication between members is impeded, for example, due to time pressure, or the necessity to hide intentions from the opposing team, teammates can't engage in explicit strategic planning but need to draw back on pre-existing knowledge, like, shared mental models. These are knowledge structures that support individuals in understanding how a system operates and what is expected in this context (Hinsz, 1995). Such models become crucial to successful team functioning because they support prediction of team members' actions and needs, allowing them to adjust their behaviors accordingly (Cannon-Bowers & Salas, 1998).

Coaches and athletes depend on good communication skills to successfully share knowledge, to convey technical and tactical ideas, to provide social and mental support (Cherubini, 2019; Ishak, 2017), and to coordinate and adjust their efforts efficiently to ultimately increase the likelihood of winning. As communication in social interaction serves the exchange of thoughts, ideas and feelings with others in a way they will understand (McKenna, 1998) good and efficient communication also implies the reduced likelihood of misunderstandings and miscommunication.

The fact that winning teams tend to communicate considerably more during competition and display more homogenous communication styles (i.e., being more in sync regarding the type of communication patterns that are used; Lausic et al., 2009) reflects the significance of team communication and shared knowledge structures in competitive sports.

According to Ishak (2017), the constant exchange between team members is a critical component in team performance with good communication being a crucial component for team success. Moreover, team sports offer a setting to improve communication efficacy.

Hence, in the present chapter we apply well-established communication theories like the four-sides-model of communication (Schulz von Thun, 1981) or the five axioms of communication (Watzlawick et al., 1967) to the sports domain as these are well suited to give guidance to coaches and players in fostering communication skills. By synthesizing a diverse body of literature, we aim at promoting an understanding of group communication in sports teams while also addressing interpersonal aspects of communication in this context. Finally, we derive applicable recommendations for coaches and athletes.

Theoretical Framework for Fostering Communication in Sports

Human communication has evolved to be highly sophisticated, which makes it unique and very useful for regulating interpersonal relations and organizing life in social groups. However, its complexity and multi-layered nature renders communication susceptible for errors and misunderstandings. Every message that is sent encompasses a multitude of informational content which can be interpreted in various ways. Difficulties may, for instance, arise through inadequate coding of messages by the sender or erroneous decoding by the recipient (Schulz von Thun, 1981). Inept wording or unclear formulation (i.e., coding) of a given information by the coach may for instance lead to misinterpretation (i.e., decoding) by the athlete. Such tendencies can be rooted in the composition of different personalities within a team and the relational dynamics among the communicators, who may differ in the way they express and interpret information (e.g., due to socio cultural differences, or language barriers; Sullivan & Callow, 2005). Thus, successful team communication depends on sharing a common code (i.e., a system of rules to exchange information) which is readily understood by everybody who is involved (Athanasios, 2005). In this vein, the

German psychologist and communication expert Friedemann Schulz von Thun (1981) proposed a model illustrating some of the versatility and challenges of human communication. Although, this early model of communication is somewhat dated and similar approaches to viewing communication have been criticized in the past for not fully capturing the complexity of the communication process (e.g., Levinson et al., 2006), it still has utility in fostering an understanding of some of the challenges of human communication. Hence, it has been argued to be a useful framework in applied sport psychology in giving heuristic guidance to coaches and players (Lau, 2020, 2023). In addition, the framework seems to be a promising starting point for structuring and initiating empirical research on interpersonal communication in sport.

The Four-Sides-Model of Communication

The four-sides-model of communication (Schulz von Thun, 1981) facilitates the analysis of social interaction and the identification of potential sources of misunderstandings. Thereby, it helps to understand communication processes and resulting interpersonal dynamics between two or more people clarifying communicational patterns and potentially preventing miscommunication. As the model comprises both verbal as well as nonverbal aspects of communication (Schulz von Thun, 2013) it is considered well suited to account for a great variety of communication behavior in the context of sports.

According to Schulz von Thun (2013), transmission of a message consists of two parts. First, information is transmitted verbally or nonverbally by the sender. It is then up to the recipient to decode and interpret the given information. Each message has four different layers that influence how the information is received: (1) The factual layer represents the raw content of a message and comprises the literal informational data of a statement; (2) The layer of self-disclosure reveals information about the sender. It contains conscious or unconscious accounts of the communicators' motives, emotions, or values; (3) The appeal-layer indicates a demand revealing what the sender of a message wants from the recipient. It can encompass desires, advice, or instructions; (4) The relationship-layer points to the nature (e.g., amicable, or collegiate) and quality (e.g., harmonious, or disharmonious) of the relationship between the sender and the recipient.

Every layer for itself can provide the recipient with a different mode of interpretation for a message. The final reception may depend on situational as well as relational factors and is highly context specific (Schulz von Thun, 2013). Thus, it is the recipient who attributes meaning to the information he or she obtains. The more room for interpretation is left by the sender, the higher is the chance for misinterpretations or misunderstandings (Schulz von Thun, 2013).

The following anecdotal scenario exemplifies the application of the model to the context of sports. During the soccer world cup in 2014 the German national coach Jogi Löw brought 22-year-old Mario Götze into the game in the 88th minute saying: "Show the world that you are better than Lionel Messi. Show that you can decide a match." The factual content of the message was: "Show everybody how good of a soccer player you are and help us win the final." Thereby, Löw revealed the motive: "I want you to play to the best of your abilities." resulting in the demand on the appeal level: "Score the decisive goal." On the relationship level Mario Götze may have heard: "I trust you and your quality as a player and I know that you can make the difference." The communication of trust may have boosted Götze's confidence, contributing to his motivation and subsequent in-game performance. However, depending on the relationship, the para- and nonverbal cues and the situational context of the interaction between coach and player, Götze could also have interpreted Löw's words differently. For instance: "Now is your last chance to prove your worth as a national player. You owe it to our fans. So, show us already that you can be useful for a change." Instead of fostering positive feelings, such an interpretation could have had a more negative impact. Thus, both interpretations

of the same message may result in completely different consequences potentially influencing the player's performance, the dynamic in the team and finally the game outcome. Acknowledging that the German victory was not solely due to Jogi Löw's choice of words his message was apparently nevertheless rather well received as Mario Götze made history when he scored the winning goal in the 113th minute granting the German national team its fourth world championship title.

This example illustrates that all sides of a message are subject to personal interpretation and its conception can therefore vary according to each individual. Thus, misunderstandings can easily occur throughout the communication process. Therefore, basic knowledge about the different facets of messages and the relation between the communicators can help analyzing communication patterns within a team. If miscommunication or conflicts arise frequently, taking a closer look at when it occurs and how certain messages are transmitted may reduce occurrence of misunderstandings and improve verbal and nonverbal exchange as well as interpersonal relations (Lau, 2020, 2023). Both are deemed beneficial for collective behavior (Ishak, 2017). While verbal communication primarily transports the content of a message (Argyle, 2005), nonverbal behavior (NVB) helps framing verbal communication and clarifying the intention behind what has been said to a certain extent (Schulz von Thun, 2013). It provides the recipient with important cues about the sender's attitudes and emotions. These cues are often elusive and therefore too intricate to be expressed verbally (DePaulo, 1992). Thereby nonverbal communication facilitates the interpretation of content, and helps determining the meaning of the verbal aspect of a message (Schulz von Thun, 2013). A friendly facial expression would for instance frame the content of a verbal message differently than a sad or angry face and therefore lead to a diverging interpretation.

Recent theorizing in sport psychology (Lau, 2020, 2023) has combined the four-sides-model of communication (Schulz von Thun, 1981) with the seminal work of Austrian-American psychologist Paul Watzlawick and colleagues who proposed five principles (pragmatic axioms) of interpersonal communication that have a high face validity for understanding communication in sports, as is outlined in the next section.

Five Axioms of Communication

As NVB is an integral part of human interaction importantly affecting interpretation of verbal messages (Argyle, 2005), psychologist and communication theorist Paul Watzlawick and colleagues (1967) have developed five **axioms** explicitly encompassing the nonverbal aspect of communication. Their work serves as an important basis in communication science helping to analyze and understand human interaction.

Axiom 1: One Cannot Not Communicate

According to Watzlawick et al. (1967), communication arises from every perceivable behavior and can happen both consciously and unconsciously. Thus, group communication starts as soon as two or more people perceive each other. Even the absence of an action is interpreted in some form by the observer and therefore conveys social information contributing to the communication process. Team members should therefore be aware that communication is a pervasive process which is not tied to active forms of interaction.

Axiom 2: Every Communication Act Has a Content and a Relationship Aspect

Similar to Schulz von Thun (1981), Watzlawick claims that what is being said is always interpreted according to the relationship of the communicators.

If a coach criticizes an athlete, it depends on the nature of their relation how the criticism is received by the athlete. If the relationship is good the athlete is more likely to accept and to interpret it as a well-intended, possibly beneficial advice. If the relationship is difficult the same criticism may be conceived less favorably increasing the chance of rejection and conflict.

Axiom 3: Communication Is Punctuated

Every communicative act can be regarded as a stimulus as well as a response or reinforcement. Punctuation of communicative sequences is subjective and may differ for each interaction partner. Thus, the perception of the beginning and development of an interaction may vary for each individual potentially resulting in a different interpretation of the same situation.

If a player is not included in the starting formation (stimulus) for a match this may lead to frustration and lack of motivation in training (response). In a conversation to resolve the situation the coach may state that the player's lack of motivation in training (stimulus) is the reason for the non-consideration (response). The individual punctuation of the same situation makes it hard to retrace cause and consequence of the interaction and may lead to opposite perspectives potentially initiating a spiraling conflict. It is important to note that communication transcends the moment (i.e., it doesn't only comprise a given conversation, but may start earlier, e.g., with a certain behavior). This event may be perceived as the start of the communication for one but not necessarily for the other interaction partner. The bigger the group the more perspectives are to be considered which increases the complexity of the communication. Given these differences, it seems beneficial to try to pinpoint one initial point of origin for a communicative situation.

Axiom 4: Communication Involves Digital and Analogous Modalities

Communication encompasses several modalities. According to Watzlawick et al. (1967) the digital modality refers to what we actually say. It mainly relates to verbal forms of interaction (e.g., spoken or written word) which follow a clear syntax. The analog modality refers to how we say something. It comprises paraverbal (e.g., speed, pitch, and tone of voice) and nonverbal (e.g., facial expressions, gestures, posture) properties providing information about inner states (e.g., attitudes and emotions).

Precise unambiguous communication relies on the congruence of both, verbal and nonverbal modalities (Argyle, 2005). If the verbal and nonverbal information sent by the communicator are incongruent, chances of misinterpretation and misunderstandings increase (Argyle, 2005). For instance, a coach verbally expressing his or her dissatisfaction with the team after a loss would be taken less seriously when displaying a happy facial expression. In cases of conflicting verbal and nonverbal information observers tend to rely more on the perceivable NVB (Burgoon et al., 2011).

Athletes and coaches should therefore be aware that both, the verbal and the nonverbal channel, importantly contribute to team communication with verbal communication transporting the factual aspect of a message and nonverbal cues providing meaning by clarifying attitudes, emotions, and intentions.

Axiom 5: Communication Can Be Symmetrical or Complementary

The characteristic of human interaction depends on whether the relationship between the communicators is based on equality or disparity. In a symmetrical relation, both interaction partners are of comparable status. Communication is generally symmetrical if interaction partners are of equal social status or share a similar level of expertise in a specific area of interest. In a complementary

constellation social status or knowledge is unequally distributed among the individuals. Differences in status often lead to one interaction partner subordinating to another. However, differences in expertise can also be used to complement one another. A head coach may for instance base decisions on years of experience in a specific field but still profit from fresh ideas provided by the young assistant coach who lacks experience but tends to think more out of the box. Both modes of communication importantly influence the dynamic of an interaction.

All five axioms are considered general communication principles that relate to verbal and non-verbal modes of communication. Both, verbal and nonverbal communication in sport teams will subsequently be focused on in more detail. Therefore, we will review several different lines of investigation within sport psychological research that are related to the outlined, broad theoretical framework on communication.

Verbal and Nonverbal Communication in Sports Teams

The goal of the next section is to integrate a highly diverse body of literature within a communication perspective and, where possible, derive some evidence-based recommendations when working in the context of sports. To this end we have divided the literature review into research on verbal and nonverbal communication in sports.

Verbal Communication

This section is intended to highlight some widely studied topics within sports (e.g., goal-setting, motivation, social support, decision making, or team cohesions) from a communication perspective. In this endeavor, we adopted the term **message-based constructs** from Ishak (2017) to describe the factual layer of the four-sides-model.

Message-Based Constructs of Sports Team Communication

Many of the efforts to manage teams and increase performance are delivered through verbal messages. The following passages intend to highlight how the factual content of some verbal messages have the potential to impact sports team behavior and or performance for the better or the worse.

Goal-Setting

An important antecedent of effective team performance are communicatively constructed goals. It is well-established that successful teams need a shared goal to effectively work together (e.g., Katzenbach & Smith, 1993; Kingston & Wilson, 2009; Williams, 2013). An important distinction within the literature on goal-setting in sports has been established between outcome goals, performance, and process goals, but also between training and competition goals, and individual and team goals (Burton & Weiss, 2008; Hall & Kerr, 2001). Hence, an evidence-informed guideline would be to set and communicate a variety of different types of goals across different time scales (e.g., Williams, 2013 for more detail). However, as studies on goal-setting interventions in sports have provided ambiguous results (Williams, 2013 for a review) and have been shown to be highly context-dependent, we refrain from giving recipe-like information on how to communicate about goals in sport teams (Ishak, 2017). Nevertheless, there seems to be a pattern in the literature on goal-setting that points to the importance of acknowledging relationship factors when setting team goals and it seems not advisable of simply having people in leading/high-status positions on a team to dictate goals to the entire teams (e.g., Williams, 2013). Thus, in team communication it

seems important to consider relationship and status of the communicators to ensure most efficient transmission of certain types of messages. A principle also indicated in Watzlawick's second and fifth axiom of communication.

Motivation and Motivational Climate

There are several lines of research within the sports literature that indicate how verbal messages can affect motivation of team members and the motivational climate of the whole team. In this respect, a seminal finding in sport psychology has been that task mastery, that is process goals, as opposed to outcome goals can lead to higher satisfaction and intrinsic motivation (Duda & Balaguer, 2007). Hence, sports teams seem well advised in communicating process goals and personal mastery (e.g., improving skills) instead of solely focusing on outcomes like winning championships.

A dominant theory within the sports motivation literature is self-determination theory (Deci & Ryan, 2011; Hagger & Chatzisarantis, 2007). This theory also has some important value concerning communication in sports teams. In a nutshell, the theory assumes that the fulfillment of three basic psychological needs (autonomy, competence, and social relatedness) leads to higher levels of intrinsic motivation with numerous positive outcomes in various sports contexts. Therefore, verbal messages that give athletes and teams the feeling of being competent, autonomous individuals and/or teams will likely be beneficial to individuals and sport teams. On the other hand, highly controlling verbal behavior by coaches is likely to undermine feelings of autonomy within teams and therefore will presumably negatively impact on athlete's motivation (Bartholomew et al., 2009; Occhino et al., 2014). In this respect, Mageau and Vallerand (2003) summarize "hands-on" strategies of how coaches' verbal behavior can increase a sports teams (intrinsic) motivation: (1) giving athletes the opportunity to choose within boundaries; (2) giving a rationale for tasks; (3) acknowledging feelings, valuing perspectives and input; (4) giving opportunities for athletes to take initiative; (5) avoid controlling comments and behaviors; and (6) reducing perception of ego-involvement. Although, these heuristic guidelines might help coaches to keep motivation high in their athletes, it is important to acknowledge that not all athletes will respond equally to verbal feedback as there may be important individual differences in how athletes want to be addressed (e.g., Horn et al., 2011). Referring back to the second axiom of communication (Watzlawick et al., 1967) and Schulz von Thun's model of communication, the influence of the relationship between the communicators needs to be considered when assessing the effect of a certain type of verbal behavior (Schulz von Thun, 2013).

Research on a coach's communication style distinguishes between a mastery-oriented climate or a performance-oriented climate (Horn et al., 2012). A mastery-oriented climate is characterized by defining success by skill-development, by valuing all players on a team, by emphasizing cooperative learning, and viewing mistakes as opportunities for learning. On the other hand, a performance-oriented or ego-involving climate emphasizes competition between team members, sanctioning of mistakes, and values higher-skilled players more than less-skilled players. Several studies (e.g., Barić & Bucik, 2009; Duda & Balaguer, 2007) have provided converging evidence for a variety of more positive outcomes (e.g., higher satisfaction, less drop-out) for a mastery-oriented climate as opposed to a performance-oriented climate.

Social Support

Starting about 20 years ago, there has been a shift regarding coaching behavior in sports that has been summarized as the "*Positive Coaching Movement*" (Denison & Avner, 2011; Thompson,

Box 23.1 The Most Memorable Messages Athletes Got From Their Coaches

A study by Cranmer et al. (2017) identified that certain types of verbal messages from a coach were particularly influential and memorable to high school athletes. In this study, messages were collected from 102 former high school athletes via open-ended questionnaires and coded for further analyses. Results indicated that three categories of messages were particularly enduring and influential in supporting athletes. The first category of messages were informational support messages telling them how to play, how to be successful, and how to relate to other team members. The second category were esteem support messages focusing on abilities, encouraging intangibles, and reinforce relationships within the team. The third category were emotional support messages that benefitted the athlete's wellbeing, offered praise, helped them improve performance, and cope with poor performance and failure. The study concludes that coaches' verbal support can take many forms, including praise, advice, instruction, and demonstrations of concern. The three identified classifications of support messages offer a promising starting point for coaches to improve their coaching effectiveness and give support to their athletes.

2003; see Box 23.1). This movement is characterized by a deemphasis of "winning at all costs" to positive development in both sports and other areas of life. In this regard the verbal behavior and communication of coaches during practice and during games has been shown to have a substantial impact on the development of athletes, particularly at younger ages (Thompson, 2003).

Although there has been a shift toward more positive communication in sports, many coaches frequently use negative messaging. One study (Walters et al., 2012) across numerous youth sports found that 35.4% of coaches' comments could be classified as positive (e.g., praise, encouragement) and 21.6% were negative, included scolding, contradicting, ridicule, and correcting without supporting. Although, this type of negative messaging might sometimes succeed to motivate athletes, research (Walters et al., 2012) has shown that negative outcomes (e.g., loss of motivation, frustration, dropping out) are correlated with this type of negative messaging. It seems plausible that such verbal behaviors may provoke a shift in social interaction from more symmetrical to more complementary forms of communication (see Axiom 5; Watzlawick et al., 1967) potentially increasing the perceived status-related social distance between coaches and athletes within a team.

Communication of Roles

A role can be described as a set of behaviors expected from a person who occupies a particular position. Research has shown that a clear communication of roles is an important component of success in sports teams (Eys et al., 2014). Various communication events occur in the process of determining and fulfilling roles. For example, the coach must determine which athlete should fill which role (e.g., field position, leader) and then communicate certain expectation toward the role. Then all players have to accept the allocated roles or negotiate the expectations (Eys et al., 2005). Studies indicate that teams with higher role clarity perform better than teams with lower role clarity (e.g., Bray et al., 2002). In addition, it was shown that teams with higher role ambiguity showed lower levels of cohesion. Hence, roles within a team should be clearly assigned and communicated between coaches and players. The communication model by Schulz von Thun and the five axioms of communication can serve as an orientation pointing out what to consider in order to communicate roles as unambiguously as possible.

All of the message-based constructs presented in this section constitute explicit forms of communication. They generally require attention of teams and team leaders, and are delivered through spoken messages and interaction (Ishak, 2017). In the following section, the focus will be placed on more implicit forms of communication which do not rely on verbal interaction but are also integral for team functioning.

Nonverbal Communication

The importance of NVB in social interaction has been acknowledged for a long time. In fact, the nonverbal share in human communication surpasses the verbal part by far as humans constantly send out nonverbal signals conveying information about inner states (e.g., attitudes, emotions, and intentions; Knapp et al., 2013). Research suggests 65%–95% of face-to-face communication to amount to NVB (Matsumoto et al., 2013). Thus, interpretation of verbal information substantially depends on ongoing processing of nonverbal cues transmitting attitudes, emotions, and intentions (see Axiom 4; Watzlawick et al., 1967) complementing and supporting interpretation of the spoken word (Argyle, 2005).

Nonverbal communication encompasses any form of signaling detached from semantics that can be perceived via sensory pathways. It includes auditory cues referred to as paralanguage (e.g., pitch, tone, speed, and rhythm of voice) and visual cues (e.g., facial expressions, gaze, gestures, and posture) also described as body language (Argyle, 2005). Furthermore, NVB comprises spatial behavior (e.g., interpersonal distance), physical features, and olfactory and haptic signals which can all be categorized as body language in a broader sense (Hess, 2016).

DePaulo (1992) emphasizes NVB to be of notable importance for interpersonal communication for various reasons: (1) In line with Watzlawick's first axiom of communication, people always convey some kind of nonverbal information. Thus, nonverbal cues are pervasive in any form of social interaction. (2) Since certain impressions are elusive and hard to express in words, they can only be conveyed nonverbally. (3) As nonverbal responses are often spontaneous and immediate, they transmit the impression of sincerity. Hence, NVB is perceived to provide rather genuine information about a respective person. (4) Nonverbal expressions are closely linked to internal states and are more accessible to the observer than to the actor. Thus, nonverbal displays can inadvertently expose emotional experience and foster specific impressions.

Even though nonverbals mostly underly some form of regulation, the attempted control is not always conscious and may therefore also unintentionally convey information to others (DePaulo, 1992). An athlete may for instance experience feelings of nervousness or anxiety immediately before a competition. This can manifest in the expression of nonverbal signals like rushing and pacing, or fidgeting gestures with hands or feet. All these behaviors may occur automatically without the athlete noticing. Such cues can readily be perceived by teammates or opponents, and may in turn affect their cognitions and subsequent actions. However, as such behaviors are semi-conscious, athletes can become aware of them and learn to control their nonverbal expressions and suppress unwanted leakage of emotional cues.

Furthermore, NVB communicates status-related social information like dominance and submissiveness which serves clarification of social hierarchies (Martens et al., 2012). A coach could, for example, add authority to a speech by adopting correspondent severe facial expressions and upright resolute posture.

In competitive sports nonverbal communication is of particular importance, as the means of verbal communication are often limited (e.g., due to the size of the court or the amount of noise in a stadium; Moesch et al., 2014). Furthermore, time constraints and other stressors cause players to increasingly rely on nonverbal rather than verbal communication (Eccles & Tran, 2012). Gaze

behavior, facial expressions, posture, gestures, spatial behavior, touch, paralanguage, and silence are mentioned to be commonly occurring for the purpose of communication and coordination among athletes and coaches (Renz & Greg, 2000). For instance, athletes use bodily signals to indicate in-game decisions and spontaneous mutual adjustments while coaches make gestures at the sidelines to communicate tactical instructions. A specific benefit of nonverbal communication in sports is that it allows for communicating play calls and tactical adjustments secretly in plain sight. Athletes frequently use predetermined gestures transmitting information to teammates (e.g., signs in baseball, play calls in basketball) while preventing opponents from grasping what is about to happen (Ishak, 2017).

Moreover, performance-related NVB informs on athletes' and coaches' emotional states. On the one hand, successful sports performance is usually followed by celebratory gestures which are often spontaneous outbursts of joy and triumph but at the same time signal strength and dominance to the observer. On the other hand, failure frequently entails shameful bodily expressions conveying impressions of weakness and submission (Tracy & Matsumoto, 2008).

Status-Related Nonverbal Displays

In sports, status-related NVBs frequently occur among athletes and coaches. This includes, for instance, the occupation of more space by adopting an erect posture with a wider stance and expanded chest – a behavior seen countless times during Cristiano Ronaldo's infamous freekick preparations. Triumphant gestures like extending the arms from the body or over the shoulders make individuals appear taller signaling confidence and superiority (Matsumoto & Hwang, 2012). Further, showing a clenched fist conveys feelings of anger or pride both of which are dominant signals found to influence the attribution of power (Tracy & Robins, 2007a). All of those NVBs evoke impressions of dominance and strength. In contrast, minimizing occupied space by adopting a more narrow stance and a slouched posture with hanging shoulders and the limbs close to the body conveys more submissive impressions.

Such nonverbals constitute a significant factor in team communication as they have been repeatedly shown to affect cognitive patterns related to sports performance (e.g., Furley & Schweizer, 2014; Furley et al., 2017). For instance, the observation of athletes' dominant nonverbal displays evokes increased confidence and expectations of success among teammates and decreased confidence and success expectancies among opponents (Furley & Schweizer, 2014). This correlation may be explained by the implicit association between dominant body language and positive attributes such as control, expertise, and power (Tracy & Robins, 2007a, 2007b). The observation of dominant body language thereby transmits the impression of an athlete being capable and in control of the situation (Furley et al., 2012) which can push teammates and support individual and team performance (Furley & Schweizer, 2020). In contrast, observation of submissive body language (e.g., slouched posture; head and gaze pointing down; limbs touching the torso; minimizing the occupied space by collapsing the body inwards, shoulders hanging to the front) implicitly linked to low social status, entails converse effects resulting in performance advantages for the opponent (Furley & Dicks, 2012). An observational study conducted by Moll et al. (2010) found dominant post-performance gestures in soccer to be associated with a deterioration of opponents' subsequent performance. More specifically, players performed significantly worse in penalty shootouts after opponents celebrated a goal by extending both arms from the body. Moreover, players extending both arms from the body while clenching both fists were significantly more likely to be in the team that ultimately won the shootout (Moll et al., 2010).

Results suggest that status-related NVB may indeed be an important factor in team communication influencing both teammates and opponents. Athletes and coaches seem well advised to openly

display dominant nonverbal signals to (1) evoke positive cognitions in team mates potentially enhancing momentum for team performance and (2) intimidate and discourage opponents decreasing their confidence in successful performance.

Tactile Communication

Another means of nonverbal communication frequently occurring in team sports is physical touch. Recent scientific advances in the domain suggest tactile communication among athletes to potentially affect team performance (e.g., Kraus et al., 2010; Moesch et al., 2013, 2014, 2018). Systematic observation in elite handball showed increased occurrence of physical touch (e.g., low five, high five, high ten, touch shoulders) during post-shot celebrations among teammates when playing well to be related to positive subsequent team performance. In contrast, showing little touch in phases when teams played well-entailed negative subsequent team performance. These findings suggest nonverbal post-shot celebrations in the form of touch after successful periods to be a good strategy for fostering psychological momentum and maintaining good team performance. However, when playing poorly high degrees of tactile team communication was related to negative subsequent team performance whereas low degrees of touch entailed more positive performance. Thus, it is suggested that timing and situational context plays an important role in the impact physical touch has within team communication.

On a more general note, studies report an increase in touching behavior during play-off games and among athletes within winning teams (Moesch et al., 2013, 2014). In a similar vein, in National Basketball Association, increased physical touch among teammates (e.g., fist bumps, high fives, chest bumps, leaping shoulder bumps, chest punches, head slaps, head grabs, low fives, high tens, full hugs, half hugs, and team huddles) in early season was found to predict greater individual and team performance later in the season (Kraus et al., 2010). More specifically, in their observational study, Kraus et al. (2010) found that the touch observed during one game early on in the season predicted enhanced team cooperation, individual, and team performance over the course of the entire season regardless of player status, preseason expectations, and early season performance.

All four studies (i.e., Kraus et al., 2010; Moesch et al., 2013, 2014, 2018) indicate tactile communication to be related to group functioning during competition. It is important to mention that these studies assessed specific patterns of physical touch in specific competitive groups (i.e., basketball and handball teams) and were conducted within a western cultural environment (i.e., Sweden and United States). Furthermore, touching behavior was only investigated within same gender teams. Cultural and gender-related differences (e.g., due to religious beliefs, or mixed gender teams) may play an important role in how touch is perceived and to what extent tactile NVB is accepted. For instance, behavioral rules concerning physical touch may differ between people of the same gender as opposed to individuals of opposite sex. Thereby, occurrence, quality, and effect of tactile NVB may be subject to change depending on the context in which it is implemented.

However, athletes and coaches should note that touch is generally a prosocial behavior highly important for bonding (Kraus et al., 2010). It communicates emotions like gratitude and sympathy and may – when applied adequately – function as positive reinforcement fostering intimacy, and encouraging compliance which is essential to cooperation within groups (Kraus et al., 2010; Moesch et al., 2014). Touch can further promote trust which is a central component of long-term cooperative bonds (Williams & Bargh, 2008). In conclusion, implementing tactile communication (e.g., hugs, high fives, fist bumps, and team huddles) in sports practice seems beneficial as it may strengthen team structure enhancing intra-team cooperation and team functioning (Kraus et al., 2010; Moesch et al., 2018).

Advantages of NVB in Sports

As team sports are characterized by high complexity, time pressure, as well as various simultaneous challenges the necessity to interact quickly and efficiently becomes evident. The swiftness of nonverbal communication, for example, an encouraging touch or gesture, is therefore of great value for team communication. Considerable benefits of NVB also lie in its capacity to transport meanings and emotions that verbal communication cannot (Burgoon et al., 2016). As certain impressions, like the reassuring feeling of a teammate's confident demeanor, are elusive, they are more difficult to verbalize than to express nonverbally (DePaulo, 1992). At the same time, nonverbal information transmitting inner states is perceived as more genuine and reliable than verbal information (Burgoon et al., 2016) which makes NVB particularly valuable for conveying impressions to observers (Argyle, 2005). Finally, an extensive body of NVB is universally comprehensible and can therefore transcend potential barriers of verbal communication (Burgoon et al., 2016) which is a great asset considering the multicultural character of modern sports.

Challenges of NVB in Sports

Despite its various benefits in team communication NVB can also be a source of misunderstandings (Burgoon et al., 2016). Especially when team members do not share the same conception about the interpretation or transmission of specific nonverbal messages (Aly, 2014). Formation of intent to produce a specific NVB is subject to cultural and situational norms (DePaulo, 1992). The overt expression of certain nonverbal displays may be encouraged in some cultures, but not in others. Moreover, there are situational factors dictating which NVBs are appropriate (DePaulo, 1992). Individuals' knowledge about such display rules determines which NVBs are taken into consideration (Furley & Schweizer, 2020). Moreover, translation of self-presentational intentions into nonverbal expressions can be inhibited by several factors. The ability to control NVBs that deviate from habitual displays may for instance be impeded by lack of expressiveness or confidence (Furley & Schweizer, 2020). An anxious or shy athlete wanting to convey signals of dominance might fail to transmit the wanted impression to his or her teammates when acting hesitantly or sending conflicting bodily signals. Therefore, clarification and synchronization of nonverbal means of communication within a team are important. This includes individual knowledge about situationally and culturally appropriate display rules as well as team members' ability to deliberately produce the desired expressions (DePaulo, 1992). However, nonverbally evoking a desired impression at will is taxing. Constraints are imposed by motivational factors, fatigue, and depletion (Matsumoto et al., 2013). Thus, under highly emotional or stressful circumstances – which frequently occur in the context of professional sports – certain expressions may be too difficult to produce or suppress at will (Furley & Schweizer, 2020) leading to failure in transmitting intended impressions or unintentionally leaking emotional cues potentially undermining self-presentational efforts (DePaulo, 1992).

Evidence-Based Recommendations for Practice

As evident from other social domains, good communication is a matter of practice. Also in team sports, the explicit focus on group and interpersonal communication skills can improve player communication competencies (Sullivan, 1993). There is an extensive body of literature claiming that good communication provides sport teams with valuable resources. This includes task-oriented resources (e.g., the exchange of tips and instructions on how to play; Widmeyer & Williams, 1991), as well as motivation prior to, during, and appraisal after competition (Hanin, 1992). Furthermore, it comprehends social resources, like, dealing with emotional messages of affection or anger (Diberardinis et al., 1983) and the acceptance of individual members

(Sullivan, 1993). Concerning the exchange of social support within teams there is a differentiation between tangible (e.g., personal assistance), emotional (e.g., listening support), and informational (e.g., task appreciation support) resources (Rosenfeld & Richman, 1997). These variables have been empirically linked to team cohesion (Widmeyer & Williams, 1991), as well as team (Hanin, 1992) and player performance (Diberardinis et al., 1983). Cherubini (2019) emphasizes that athletes benefit from harmonious and positive communication styles transmitting information with respect, empathy, honesty, and brevity. It can facilitate positive affect, self-determination, on-task behavior, and self-regulation which fosters athletes' potential to thrive both in and outside of the sporting context (Cherubini, 2019).

Also, nonverbal communication can be practiced to be more precise and efficient (Hackfort & Schlattmann, 2002). For athletes, video-based behavioral training to develop appropriate self-presentation has been suggested in the past (Hackfort & Schlattmann, 2002). The plethora of visual data available specifically in professional sports offers extensive possibilities to base NVB training on. Moreover, given that body language extends to every area of social interaction, training awareness, interpretation, and nonverbal expression is possible in specific sporting encounters as well as in general social interaction. However, repeated and deliberate practice is necessary to consolidate acquired abilities (Furley & Schweizer, 2020).

Conclusion

Considering the abundance of variables and relations within a sports team, it is not possible to fully control all aspects of social interaction while performing. However, good communication is an important pillar providing structure and supporting athlete satisfaction (Ishak, 2017). The verbal share in team communication comprehends essential processes like goal-setting, motivation, social support, sharing of knowledge, decision making, and clarification of roles (Ishak, 2017). The nonverbal portion supports these processes to a great extent (Burgoon et al., 2016) and constitutes a major factor in the communication of signals related to emotions and social status also affecting impression formation (Furley & Schweizer, 2020). Hence, fostering the synergy of well-functioning verbal and nonverbal modes of interaction in sport teams is likely to be a good way to advance team coordination, overall performance (Eccles & Tran, 2012) and enhance the sporting experience (Cherubini, 2019).

Further Readings

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Glossary

Axiom A statement accepted as an established principle or self-evident truth which is used as the basis for further reasoning or inference.

Message-based constructs Explicit forms of team interaction that are based on the verbal exchange of information, feedback, and/or knowledge.

Nonverbal communication The transfer of information by means of body language (e.g., facial expressions, gaze, gestures, and posture) and/or paralanguage (e.g., pitch, tone, speed, and rhythm of voice).

Verbal communication The conveyance of a message by means of spoken or written words.

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